

Knowledge Distillation Evaluation Report

This report is generated from JSON files in [results/evaluation](#).

Raw accuracies are analyzed separately per dataset. The global section only uses normalized comparisons such as improvement over the dataset baseline, improvement over the matching teacher, and cost ratios.

Plot conventions:

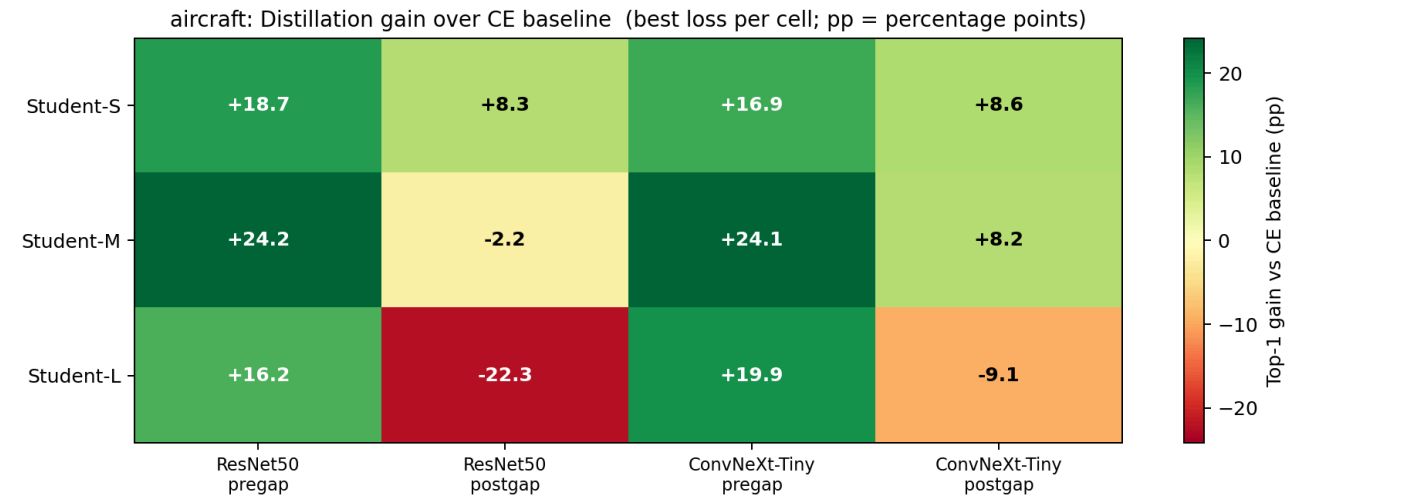
- colour = student architecture (blue S-S, orange S-M, green S-L);
- overlay = distillation target (pre-GAP, post-GAP);
- edge = loss (black edge MSE+CE, no edge MSE-only);
- shape:
 - ★ = teacher;
 - ◆ = CE baseline;
 - = ResNet-50;
 - = ConvNeXt-Tiny;

Executive Summary By Dataset

- **aircraft**: best student is **convnext_tiny + student_l + pregap + mse_ce** at 55.566% top-1 (19.922 pp vs matching baseline; 77.486% GFLOPs saved vs teacher).
- **food101**: best student is **convnext_tiny + student_l + pregap + mse_ce** at 68.376% top-1 (-1.857 pp vs matching baseline; 77.486% GFLOPs saved vs teacher).

Dataset: aircraft

- Best student CE baseline: **student_l** with 35.644% top-1.
- Best teacher classifier: **convnext_tiny** with 58.686% top-1.
- Best distilled student: **convnext_tiny + student_l + pregap + mse_ce** with 55.566% top-1 (19.922 pp vs matching baseline, -3.120 pp vs its teacher).



Evaluation Results

model_type	teacher	student	target	loss_name	top1	top5	loss	mse	top1_minus_baseline	top1_minus_teacher	cost_ratio
baseline		student_l		ce	35.644	69.337	2.422		0.000		162086
baseline		student_m		ce	16.472	44.644	3.376		0.000		414596
baseline		student_s		ce	7.231	23.252	4.168		0.000		106372
distilled	convnext_tiny	student_l	postgap	mse	15.602	49.565	3.529	0.222	-20.042	-43.084	263258
distilled	convnext_tiny	student_l	postgap	mse_ce	26.553	61.386	2.757	0.386	-9.091	-32.133	263258
distilled	convnext_tiny	student_l	pregap	mse	24.842	62.106	2.759	0.998	-10.801	-33.843	213952
distilled	convnext_tiny	student_l	pregap	mse_ce	55.566	86.019	1.544	1.458	19.922	-3.120	213952
distilled	convnext_tiny	student_m	postgap	mse	11.221	36.004	4.122	0.244	-5.251	-47.465	125530
distilled	convnext_tiny	student_m	postgap	mse_ce	24.692	58.536	2.917	0.390	8.221	-33.993	125530
distilled	convnext_tiny	student_m	pregap	mse	8.851	29.043	4.155	1.197	-7.621	-49.835	860548
distilled	convnext_tiny	student_m	pregap	mse_ce	40.564	76.748	2.128	1.581	24.092	-18.122	860548

model_type	teacher	student	target	loss_name	top1	top5	loss	mse	top1_minus_baseline	top1_minus_teacher	cost_per_epoch
distilled	convnext_tiny	student_s	postgap	mse	7.681	26.973	4.716	0.263	0.450	-51.005	861572
distilled	convnext_tiny	student_s	postgap	mse_ce	15.812	41.704	3.462	0.448	8.581	-42.874	861572
distilled	convnext_tiny	student_s	pregap	mse	3.600	13.081	5.476	1.376	-3.630	-55.086	515972
distilled	convnext_tiny	student_s	pregap	mse_ce	24.152	57.576	2.937	1.658	16.922	-34.533	515972
distilled	resnet50	student_l	postgap	mse	13.381	43.474	3.450	0.086	-22.262	-40.354	440102
distilled	resnet50	student_l	postgap	mse_ce	6.961	21.452	4.187	0.165	-28.683	-46.775	440102
distilled	resnet50	student_l	pregap	mse	39.004	75.638	2.170	1.045	3.360	-14.731	440000
distilled	resnet50	student_l	pregap	mse_ce	51.875	83.198	1.760	1.519	16.232	-1.860	440000
distilled	resnet50	student_m	postgap	mse	9.661	34.233	3.907	0.088	-6.811	-44.074	295821
distilled	resnet50	student_m	postgap	mse_ce	14.281	40.384	3.534	0.164	-2.190	-39.454	295821
distilled	resnet50	student_m	pregap	mse	26.493	62.166	2.765	1.133	10.021	-27.243	295718
distilled	resnet50	student_m	pregap	mse_ce	40.654	75.128	2.194	1.524	24.182	-13.081	295718
distilled	resnet50	student_s	postgap	mse	8.881	31.593	4.015	0.089	1.650	-44.854	253171
distilled	resnet50	student_s	postgap	mse_ce	15.482	42.634	3.477	0.161	8.251	-38.254	253171
distilled	resnet50	student_s	pregap	mse	10.291	34.053	3.778	1.233	3.060	-43.444	253069
distilled	resnet50	student_s	pregap	mse_ce	25.893	58.986	2.917	1.509	18.662	-27.843	253069
teacher	convnext_tiny				58.686	87.999	1.431			0.000	278954
teacher	resnet50				53.735	82.658	1.846			0.000	237129

Question 1: Which Teacher Transfers Best?

Answer: **resnet50** transfers best on this dataset by mean distilled Top-1 (21.905%).

teacher	mean	max	min	count
resnet50	21.905	51.875	6.961	12
convnext_tiny	21.595	55.566	3.600	12

Best configuration found for each teacher:

teacher	best_student	best_target	best_loss	best_top1	best_top5	best_minus_baseline	best_minus_teacher
convnext_tiny	student_l	pregap	mse_ce	55.566	86.019	19.922	-3.120
resnet50	student_l	pregap	mse_ce	51.875	83.198	16.232	-1.860

Question 2: What Should The Student Predict?

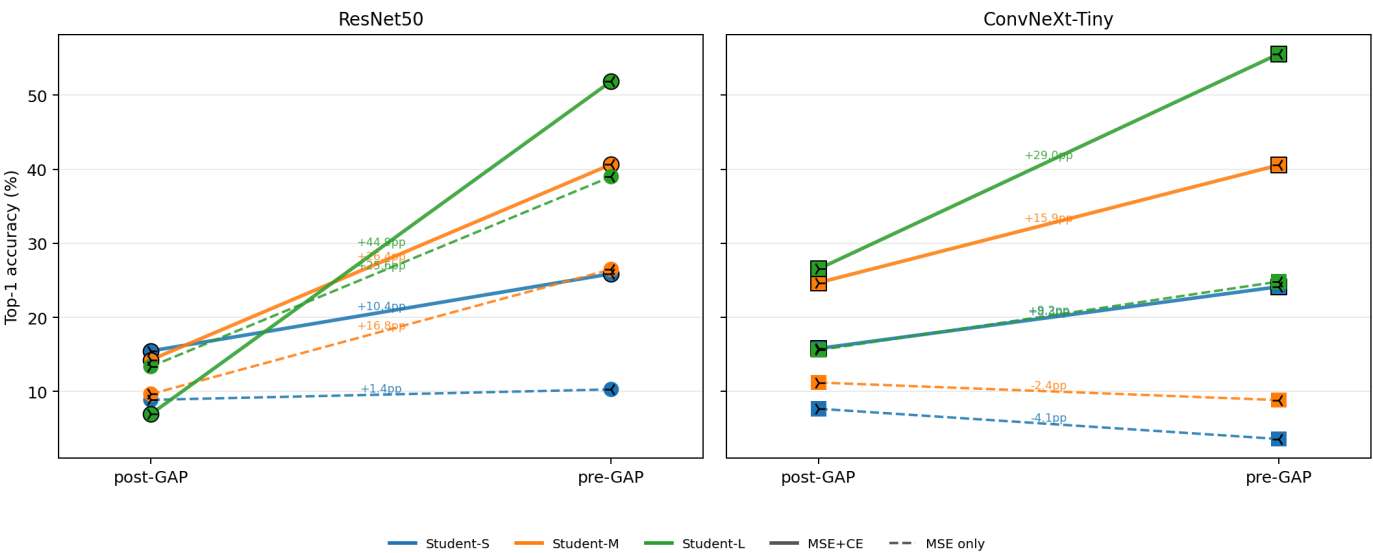
Answer: **pregap** performs best on this dataset by mean Top-1; this corresponds to pre-GAP feature map with convolutional predictor.

target	mean	max	min	count
pregap	29.315	55.566	3.600	12
postgap	14.184	26.553	6.961	12

Best configuration found for each target:

target	best_teacher	best_student	best_loss	best_top1	best_top5	best_minus_baseline	best_minus_teacher
pregap	convnext_tiny	student_l	mse_ce	55.566	86.019	19.922	-3.120
postgap	convnext_tiny	student_l	mse_ce	26.553	61.386	-9.091	-32.133

aircraft: Distillation target comparison — post-GAP vs pre-GAP



Question 3: What Is The Best Student Architecture?

Answer: **student_l** (Student-L (Conv32 → Conv64 → Conv128 → Conv256 → Conv512)) achieves the best mean distilled Top-1 (29.223%) on this dataset.

student	mean	max	min	count
student_l	29.223	55.566	6.961	8
student_m	22.052	40.654	8.851	8
student_s	13.974	25.893	3.600	8

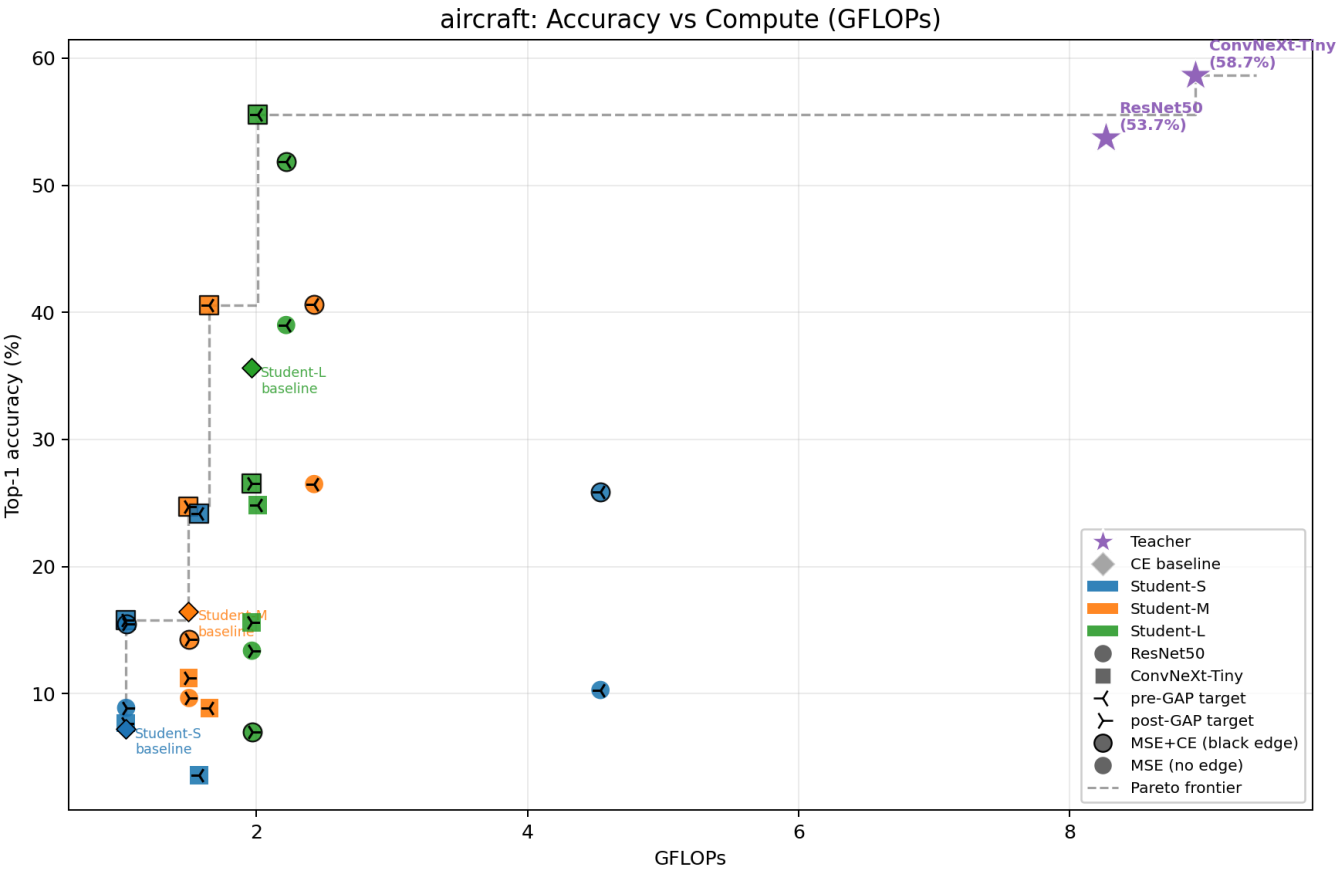
Best configuration found for each student architecture:

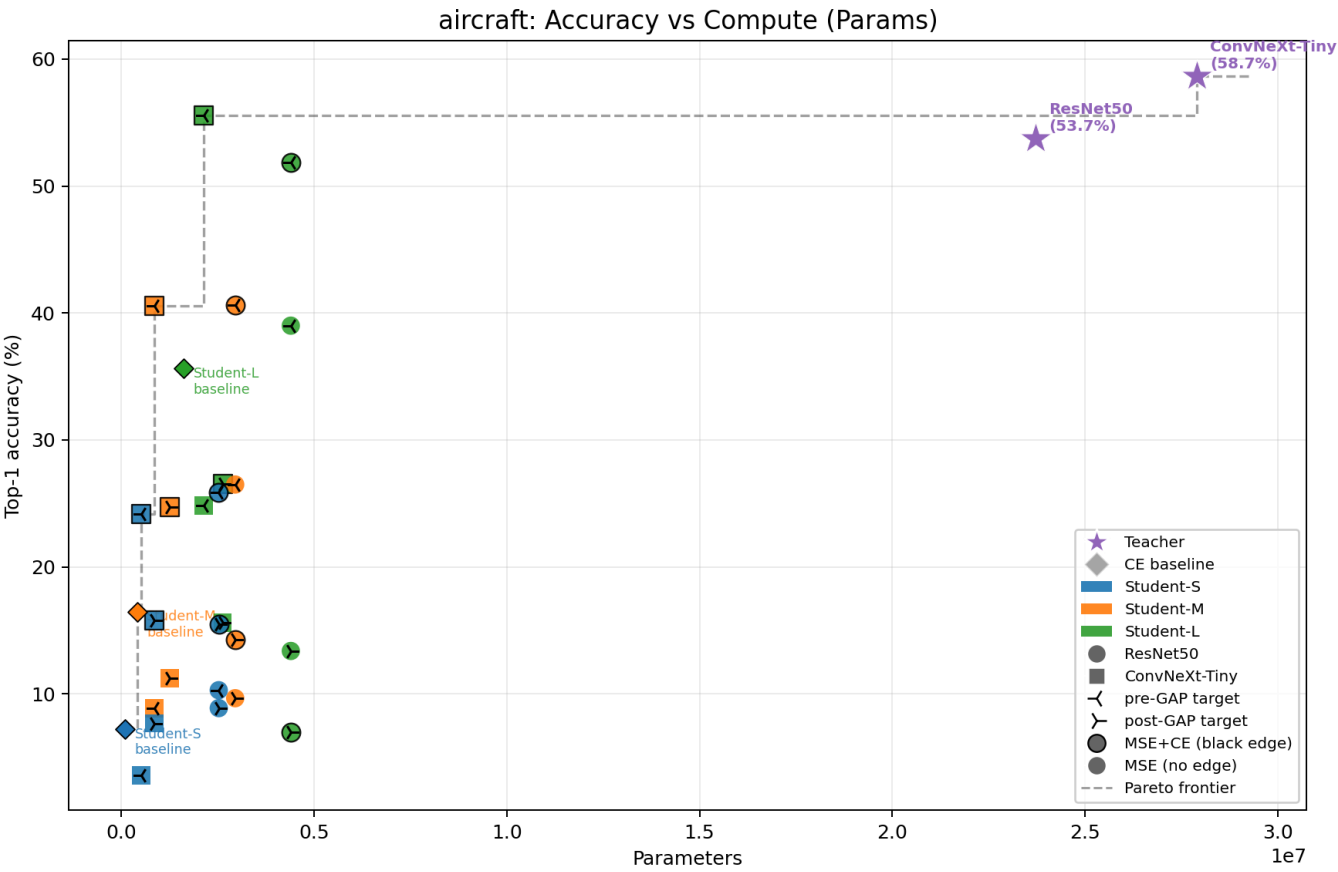
student	best_teacher	best_target	best_loss	best_top1	best_top5	best_minus_baseline	best_minus_teacher
student_l	convnext_tiny	pregap	mse_ce	55.566	86.019	19.922	-3.120
student_m	resnet50	pregap	mse_ce	40.654	75.128	24.182	-13.081
student_s	resnet50	pregap	mse_ce	25.893	58.986	18.662	-27.843

Architecture ranking with cost savings relative to the matching teacher:

teacher	student	target	loss_name	top1	top5	cost_params	cost_gflops	params_saved_vs_teacher_pct	gflops_saved_vs_teach
convnext_tiny	student_l	pregap	mse_ce	55.566	86.019	2139524	2.010	92.330	77.486
resnet50	student_l	pregap	mse_ce	51.875	83.198	4400004	2.219	81.445	73.146
resnet50	student_m	pregap	mse_ce	40.654	75.128	2957188	2.426	87.529	70.646
convnext_tiny	student_m	pregap	mse_ce	40.564	76.748	860548	1.653	96.915	81.480
resnet50	student_l	pregap	mse	39.004	75.638	4400004	2.219	81.445	73.146
convnext_tiny	student_l	postgap	mse_ce	26.553	61.386	2632580	1.963	90.563	78.005
resnet50	student_m	pregap	mse	26.493	62.166	2957188	2.426	87.529	70.646
resnet50	student_s	pregap	mse_ce	25.893	58.986	2530692	4.538	89.328	45.087
convnext_tiny	student_l	pregap	mse	24.842	62.106	2139524	2.010	92.330	77.486
convnext_tiny	student_m	postgap	mse_ce	24.692	58.536	1255300	1.500	95.500	83.198
convnext_tiny	student_s	pregap	mse_ce	24.152	57.576	515972	1.577	98.150	82.331
convnext_tiny	student_s	postgap	mse_ce	15.812	41.704	861572	1.036	96.911	88.397
convnext_tiny	student_l	postgap	mse	15.602	49.565	2632580	1.963	90.563	78.005
resnet50	student_s	postgap	mse_ce	15.482	42.634	2531716	1.039	89.323	87.426
resnet50	student_m	postgap	mse_ce	14.281	40.384	2958212	1.503	87.525	81.809
resnet50	student_l	postgap	mse	13.381	43.474	4401028	1.967	81.440	76.197
convnext_tiny	student_m	postgap	mse	11.221	36.004	1255300	1.500	95.500	83.198

teacher	student	target	loss_name	top1	top5	cost_params	cost_gflops	params_saved_vs_teacher_pct	gflops_saved_vs_teach
resnet50	student_s	pregap	mse	10.291	34.053	2530692	4.538	89.328	45.087
resnet50	student_m	postgap	mse	9.661	34.233	2958212	1.503	87.525	81.809
resnet50	student_s	postgap	mse	8.881	31.593	2531716	1.039	89.323	87.426
convnext_tiny	student_m	pregap	mse	8.851	29.043	860548	1.653	96.915	81.480
convnext_tiny	student_s	postgap	mse	7.681	26.973	861572	1.036	96.911	88.397
resnet50	student_l	postgap	mse_ce	6.961	21.452	4401028	1.967	81.440	76.197
convnext_tiny	student_s	pregap	mse	3.600	13.081	515972	1.577	98.150	82.331





Question 4: What Loss Function Should We Use?

Answer: `mse_ce` performs best on this dataset by mean distilled Top-1 (28.540%).

loss_name	mean	max	min	count
mse_ce	28.540	55.566	6.961	12
mse	14.959	39.004	3.600	12

Best configuration found for each loss:

loss_name	best_teacher	best_student	best_target	best_top1	best_top5	best_minus_baseline	best_minus_teacher
mse_ce	convnext_tiny	student_l	pregap	55.566	86.019	19.922	-3.120
mse	resnet50	student_l	pregap	39.004	75.638	3.360	-14.731

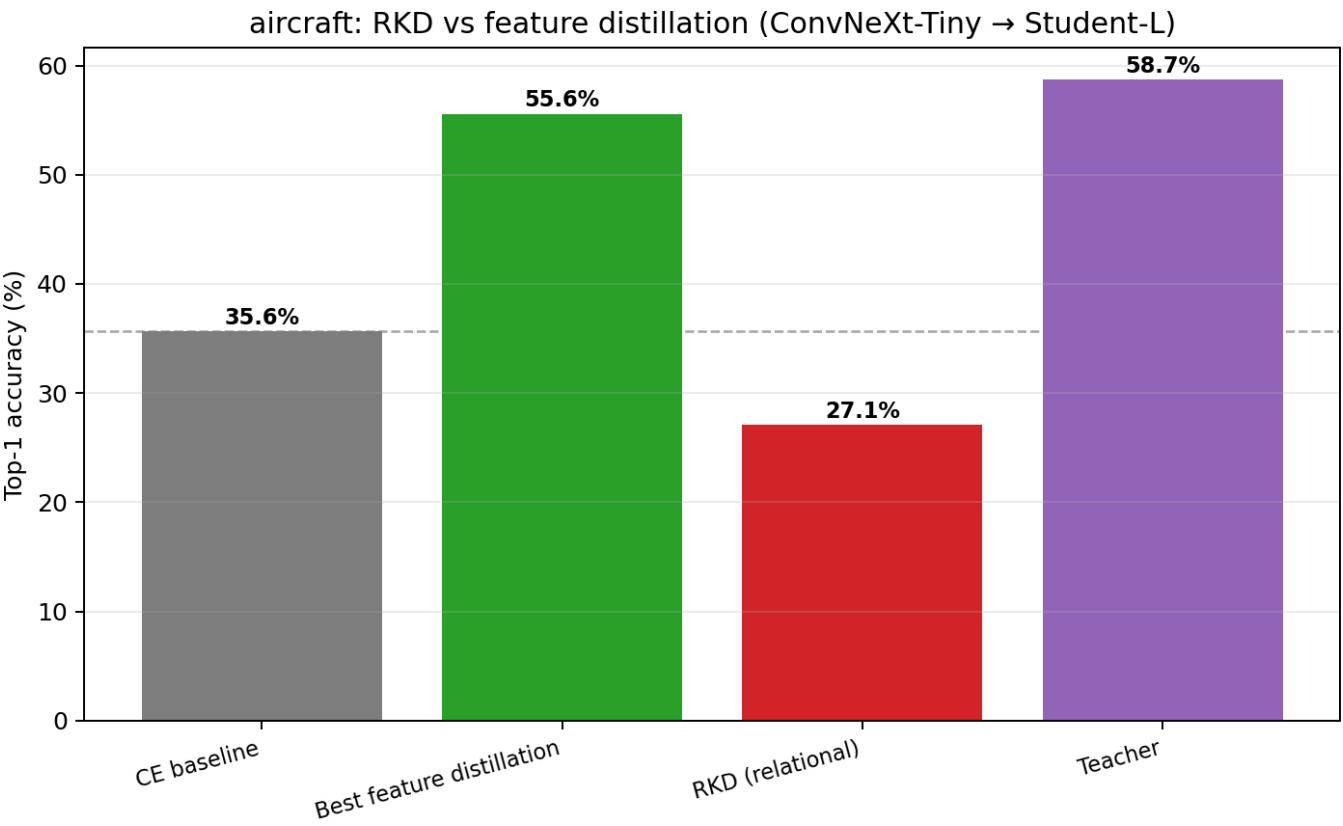
Question 5: How Does Relational Knowledge Distillation Compare?

This question evaluates Relational Knowledge Distillation (Park et al., 2019) as a literature-based extension. Unlike feature distillation, RKD keeps the student's own classifier and transfers the *relations* between examples, namely normalized pairwise distances and triplet angles computed over the pooled embeddings, rather than forcing the student to match the teacher's feature values directly.

Because these relations are computed within each model's embedding space, the teacher and student embedding dimensions do not need to match. RKD is trained on the same `convnext_tiny` + `student_l` pair as the strongest feature-distillation configuration, allowing a direct comparison between the two distillation paradigms.

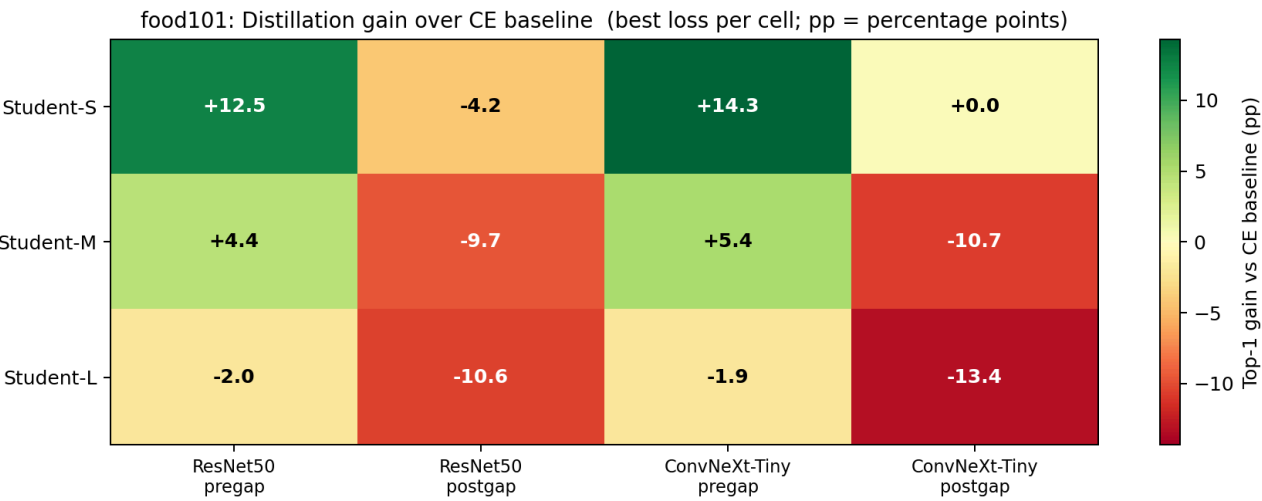
Answer: RKD reaches 27.063% Top-1, which trails the best feature-distillation result for the same pair by 28.503 pp and is 8.581 pp below the CE baseline.

method	top1	top5	top1_minus_baseline	top1_minus_teacher	cost_params	cost_gflops	gflops_saved_vs_teacher_pct
CE baseline	35.644	69.337	0.000		1620868	1.961	
Best feature distillation	55.566	86.019	19.922	-3.120	2139524	2.010	77.486
RKD (relational)	27.063	62.226	-8.581	-31.623	1620868	1.961	78.028
Teacher	58.686	87.999		0.000	27895492	8.927	0.000



Dataset: food101

- Best student CE baseline: `student_l` with 70.234% top-1.
- Best teacher classifier: `convnext_tiny` with 82.923% top-1.
- Best distilled student: `convnext_tiny` + `student_l` + `pregap` + `mse_ce` with 68.376% top-1 (-1.857 pp vs matching baseline, -14.547 pp vs its teacher).



Evaluation Results

model_type	teacher	student	target	loss_name	top1	top5	loss	mse	top1_minus_baseline	top1_minus_teacher	cost_pp
baseline		student_l	ce		70.234	90.388	1.123		0.000		162138
baseline		student_m	ce		58.717	83.295	1.635		0.000		414853
baseline		student_s	ce		39.687	68.107	2.500		0.000		106501
distilled	convnext_tiny	student_l	postgap	mse	56.808	84.250	1.721	0.033	-13.426	-26.115	263334
distilled	convnext_tiny	student_l	postgap	mse_ce	44.170	72.463	2.227	0.048	-26.063	-38.752	263334
distilled	convnext_tiny	student_l	pregap	mse	65.248	89.077	1.302	0.267	-4.986	-17.675	214029
distilled	convnext_tiny	student_l	pregap	mse_ce	68.376	89.370	1.235	0.342	-1.857	-14.547	214029

model_type	teacher	student	target	loss_name	top1	top5	loss	mse	top1_minus_baseline	top1_minus_teacher	cost_percent
distilled	convnext_tiny	student_m	postgap	mse	48.024	77.386	2.117	0.035	-10.693	-34.899	125606
distilled	convnext_tiny	student_m	postgap	mse_ce	41.386	69.980	2.383	0.048	-17.331	-41.537	125606
distilled	convnext_tiny	student_m	pregap	mse	52.622	80.816	1.841	0.282	-6.095	-30.301	861317
distilled	convnext_tiny	student_m	pregap	mse_ce	64.087	86.990	1.383	0.339	5.370	-18.836	861317
distilled	convnext_tiny	student_s	postgap	mse	34.436	64.432	2.836	0.038	-5.251	-48.487	862341
distilled	convnext_tiny	student_s	postgap	mse_ce	39.703	68.186	2.467	0.048	0.016	-43.220	862341
distilled	convnext_tiny	student_s	pregap	mse	36.408	67.141	2.575	0.294	-3.279	-46.515	516741
distilled	convnext_tiny	student_s	pregap	mse_ce	54.016	80.372	1.809	0.327	14.329	-28.907	516741
distilled	resnet50	student_l	postgap	mse	59.604	85.390	1.593	0.069	-10.630	-19.794	440307
distilled	resnet50	student_l	postgap	mse_ce	32.412	61.687	2.771	0.114	-37.822	-46.986	440307
distilled	resnet50	student_l	pregap	mse	68.246	90.016	1.217	0.843	-1.988	-11.152	440205
distilled	resnet50	student_l	pregap	mse_ce	67.350	88.935	1.248	1.041	-2.883	-12.048	440205
distilled	resnet50	student_m	postgap	mse	49.026	77.770	2.056	0.074	-9.691	-30.372	296026
distilled	resnet50	student_m	postgap	mse_ce	35.727	64.824	2.625	0.114	-22.990	-43.671	296026
distilled	resnet50	student_m	pregap	mse	53.984	80.962	1.817	0.916	-4.733	-25.414	295923
distilled	resnet50	student_m	pregap	mse_ce	63.089	86.285	1.437	1.050	4.372	-16.309	295923
distilled	resnet50	student_s	postgap	mse	35.358	64.685	2.815	0.079	-4.329	-44.040	253376
distilled	resnet50	student_s	postgap	mse_ce	35.489	64.685	2.786	0.114	-4.198	-43.909	253376
distilled	resnet50	student_s	pregap	mse	36.388	65.200	2.662	0.965	-3.299	-43.010	253274
distilled	resnet50	student_s	pregap	mse_ce	52.154	78.749	1.900	1.048	12.467	-27.244	253274
teacher	convnext_tiny				82.923	96.281	0.718			0.000	278962
teacher	resnet50				79.398	94.638	0.894			0.000	237149

Question 1: Which Teacher Transfers Best?

Answer: **convnext_tiny** transfers best on this dataset by mean distilled Top-1 (50.440%).

teacher	mean	max	min	count
convnext_tiny	50.440	68.376	34.436	12
resnet50	49.069	68.246	32.412	12

Best configuration found for each teacher:

teacher	best_student	best_target	best_loss	best_top1	best_top5	best_minus_baseline	best_minus_teacher
convnext_tiny	student_l	pregap	mse_ce	68.376	89.370	-1.857	-14.547
resnet50	student_l	pregap	mse	68.246	90.016	-1.988	-11.152

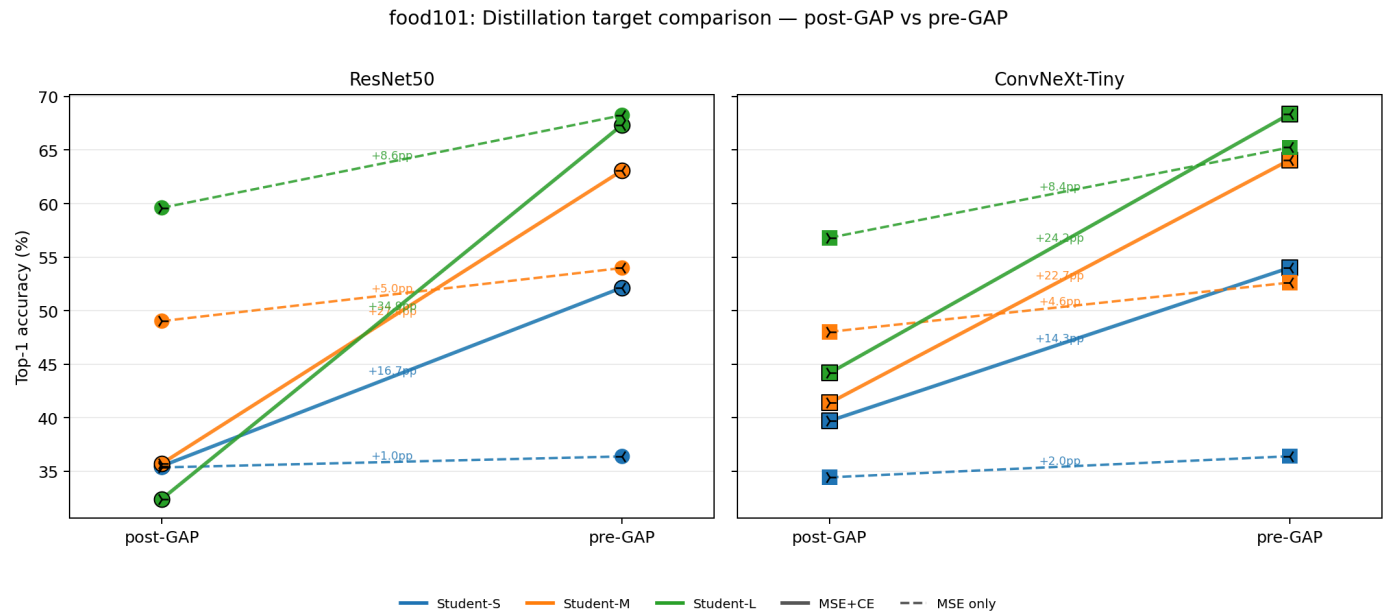
Question 2: What Should The Student Predict?

Answer: **pregap** performs best on this dataset by mean Top-1; this corresponds to pre-GAP feature map with convolutional predictor.

target	mean	max	min	count
pregap	56.831	68.376	36.388	12
postgap	42.679	59.604	32.412	12

Best configuration found for each target:

target	best_teacher	best_student	best_loss	best_top1	best_top5	best_minus_baseline	best_minus_teacher
pregap	convnext_tiny	student_l	mse_ce	68.376	89.370	-1.857	-14.547
postgap	resnet50	student_l	mse	59.604	85.390	-10.630	-19.794



Question 3: What Is The Best Student Architecture?

Answer: **student_l** (Student-L (Conv32 → Conv64 → Conv128 → Conv256 → Conv512)) achieves the best mean distilled Top-1 (57.777%) on this dataset.

student	mean	max	min	count
student_l	57.777	68.376	32.412	8
student_m	50.993	64.087	35.727	8
student_s	40.494	54.016	34.436	8

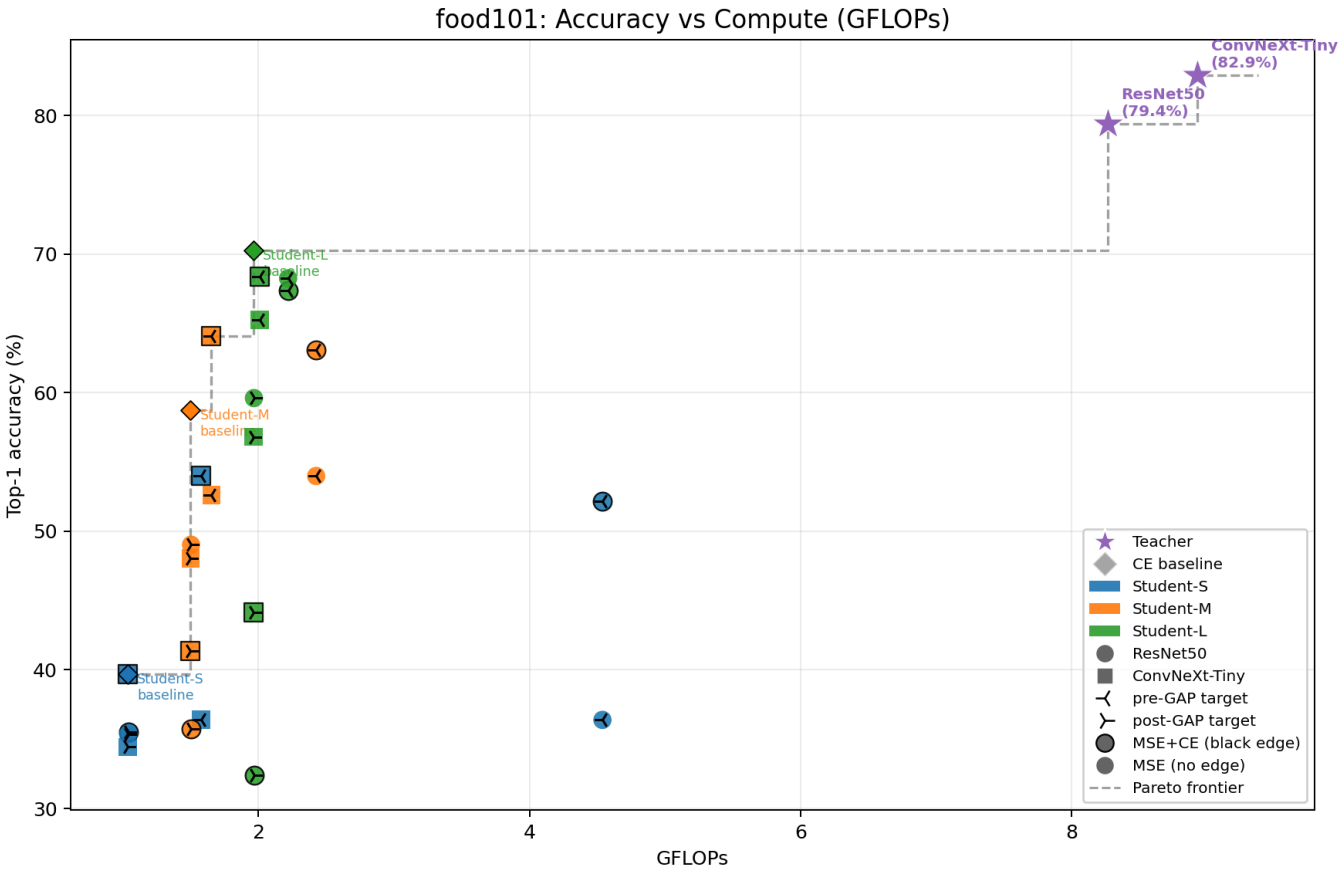
Best configuration found for each student architecture:

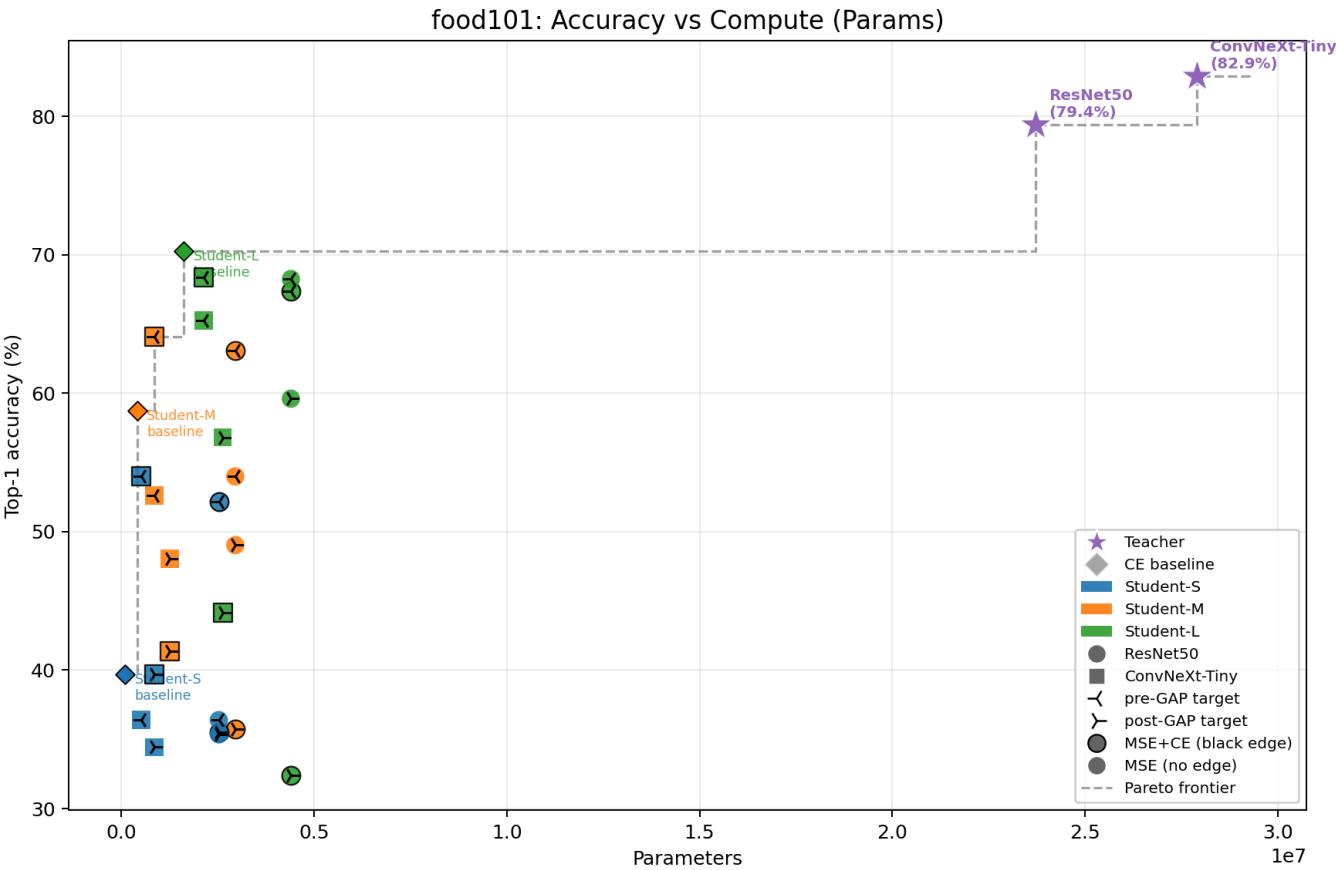
student	best_teacher	best_target	best_loss	best_top1	best_top5	best_minus_baseline	best_minus_teacher
student_l	convnext_tiny	pregap	mse_ce	68.376	89.370	-1.857	-14.547
student_m	convnext_tiny	pregap	mse_ce	64.087	86.990	5.370	-18.836
student_s	convnext_tiny	pregap	mse_ce	54.016	80.372	14.329	-28.907

Architecture ranking with cost savings relative to the matching teacher:

teacher	student	target	loss_name	top1	top5	cost_params	cost_gflops	params_saved_vs_teacher_pct	gflops_saved_vs_teach
convnext_tiny	student_l	pregap	mse_ce	68.376	89.370	2140293	2.010	92.328	77.486
resnet50	student_l	pregap	mse	68.246	90.016	4402053	2.219	81.438	73.146
resnet50	student_l	pregap	mse_ce	67.350	88.935	4402053	2.219	81.438	73.146
convnext_tiny	student_l	pregap	mse	65.248	89.077	2140293	2.010	92.328	77.486
convnext_tiny	student_m	pregap	mse_ce	64.087	86.990	861317	1.653	96.912	81.480
resnet50	student_m	pregap	mse_ce	63.089	86.285	2959237	2.426	87.522	70.646
resnet50	student_l	postgap	mse	59.604	85.390	4403077	1.967	81.433	76.197
convnext_tiny	student_l	postgap	mse	56.808	84.250	2633349	1.963	90.560	78.005
convnext_tiny	student_s	pregap	mse_ce	54.016	80.372	516741	1.577	98.148	82.331
resnet50	student_m	pregap	mse	53.984	80.962	2959237	2.426	87.522	70.646
convnext_tiny	student_m	pregap	mse	52.622	80.816	861317	1.653	96.912	81.480
resnet50	student_s	pregap	mse_ce	52.154	78.749	2532741	4.538	89.320	45.087
resnet50	student_m	postgap	mse	49.026	77.770	2960261	1.503	87.517	81.809
convnext_tiny	student_m	postgap	mse	48.024	77.386	1256069	1.500	95.497	83.198
convnext_tiny	student_l	postgap	mse_ce	44.170	72.463	2633349	1.963	90.560	78.005
convnext_tiny	student_m	postgap	mse_ce	41.386	69.980	1256069	1.500	95.497	83.198
convnext_tiny	student_s	postgap	mse_ce	39.703	68.186	862341	1.036	96.909	88.397

teacher	student	target	loss_name	top1	top5	cost_params	cost_gflops	params_saved_vs_teacher_pct	gflops_saved_vs_teach
convnext_tiny	student_s	pregap	mse	36.408	67.141	516741	1.577	98.148	82.331
resnet50	student_s	pregap	mse	36.388	65.200	2532741	4.538	89.320	45.087
resnet50	student_m	postgap	mse_ce	35.727	64.824	2960261	1.503	87.517	81.809
resnet50	student_s	postgap	mse_ce	35.489	64.685	2533765	1.039	89.316	87.426
resnet50	student_s	postgap	mse	35.358	64.685	2533765	1.039	89.316	87.426
convnext_tiny	student_s	postgap	mse	34.436	64.432	862341	1.036	96.909	88.397
resnet50	student_l	postgap	mse_ce	32.412	61.687	4403077	1.967	81.433	76.197





Question 4: What Loss Function Should We Use?

Answer: `mse_ce` performs best on this dataset by mean distilled Top-1 (49.830%).

loss_name	mean	max	min	count
mse_ce	49.830	68.376	32.412	12
mse	49.679	68.246	34.436	12

Best configuration found for each loss:

loss_name	best_teacher	best_student	best_target	best_top1	best_top5	best_minus_baseline	best_minus_teacher
mse_ce	convnext_tiny	student_l	pregap	68.376	89.370	-1.857	-14.547
mse	resnet50	student_l	pregap	68.246	90.016	-1.988	-11.152

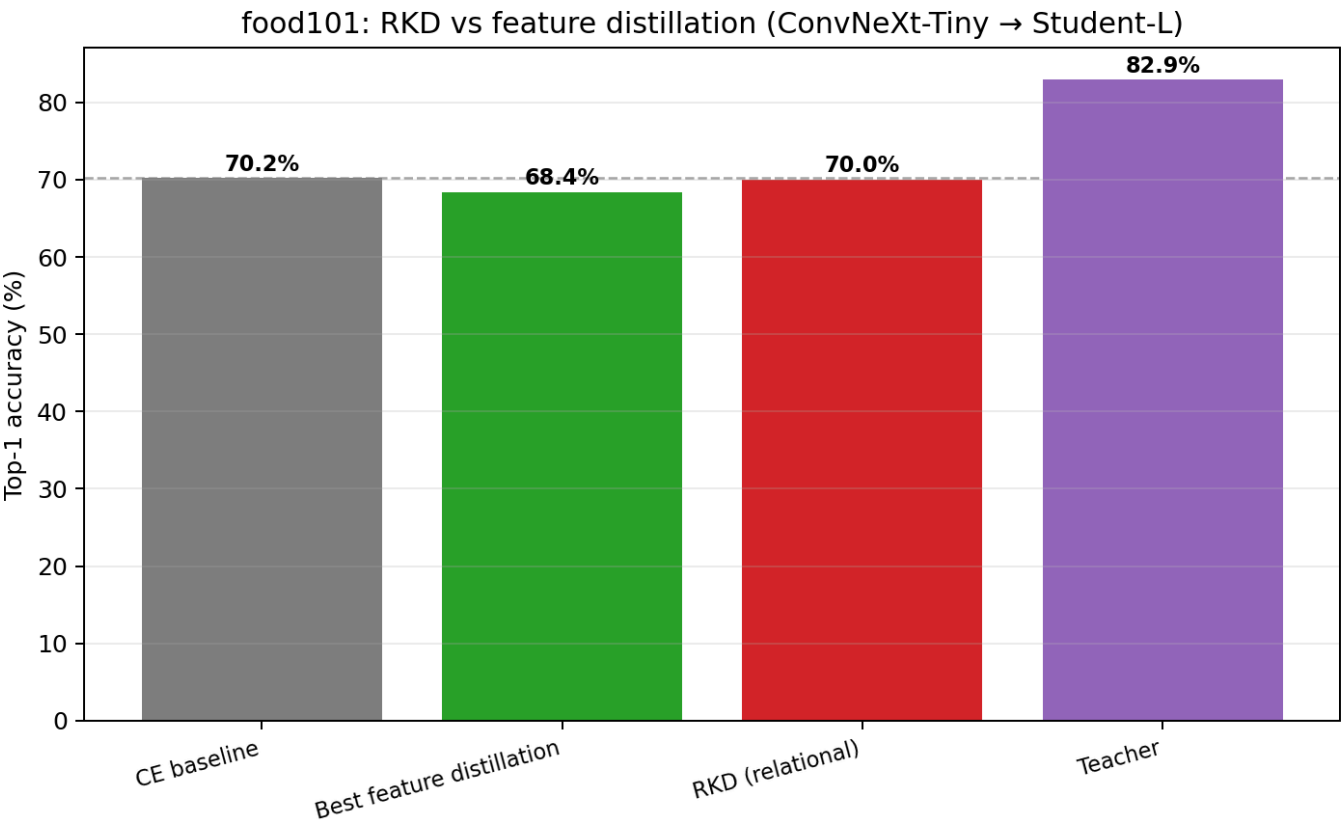
Question 5: How Does Relational Knowledge Distillation Compare?

This question evaluates Relational Knowledge Distillation (Park et al., 2019) as a literature-based extension. Unlike feature distillation, RKD keeps the student's own classifier and transfers the *relations* between examples, namely normalized pairwise distances and triplet angles computed over the pooled embeddings, rather than forcing the student to match the teacher's feature values directly.

Because these relations are computed within each model's embedding space, the teacher and student embedding dimensions do not need to match. RKD is trained on the same `convnext_tiny` + `student_l` pair as the strongest feature-distillation configuration, allowing a direct comparison between the two distillation paradigms.

Answer: RKD reaches 70.016% Top-1, which outperforms the best feature-distillation result for the same pair by 1.640 pp and is 0.218 pp below the CE baseline.

method	top1	top5	top1_minus_baseline	top1_minus_teacher	cost_params	cost_gflops	gflops_saved_vs_teacher_pct
CE baseline	70.234	90.388	0.000		1621381	1.961	
Best feature distillation	68.376	89.370	-1.857	-14.547	2140293	2.010	77.486
RKD (relational)	70.016	89.929	-0.218	-12.907	1621381	1.961	78.028
Teacher	82.923	96.281		0.000	27896261	8.927	0.000



Global Overview With Normalized Comparisons

Best distilled student per dataset:

dataset	teacher	student	target	loss_name	top1	top1_minus_baseline	top1_minus_teacher	cost_params	cost_gflops	params_s
aircraft	convnext_tiny	student_l	pregap	mse_ce	55.566	19.922	-3.120	2139524	2.010	92.330
food101	convnext_tiny	student_l	pregap	mse_ce	68.376	-1.857	-14.547	2140293	2.010	92.328

Global teacher overview using normalized deltas:

teacher	top1_minus_baseline	top1_minus_teacher	gflops_saved_vs_teacher_pct	params_saved_vs_teacher_pct
convnext_tiny	-1.980	-34.787	81.816	95.060
resnet50	-2.510	-31.080	72.385	86.095

Global target overview using normalized deltas:

target	top1_minus_baseline	top1_minus_teacher	gflops_saved_vs_teacher_pct	params_saved_vs_teacher_pct
pregap	5.076	-25.612	71.696	90.947
postgap	-9.566	-40.254	82.505	90.208

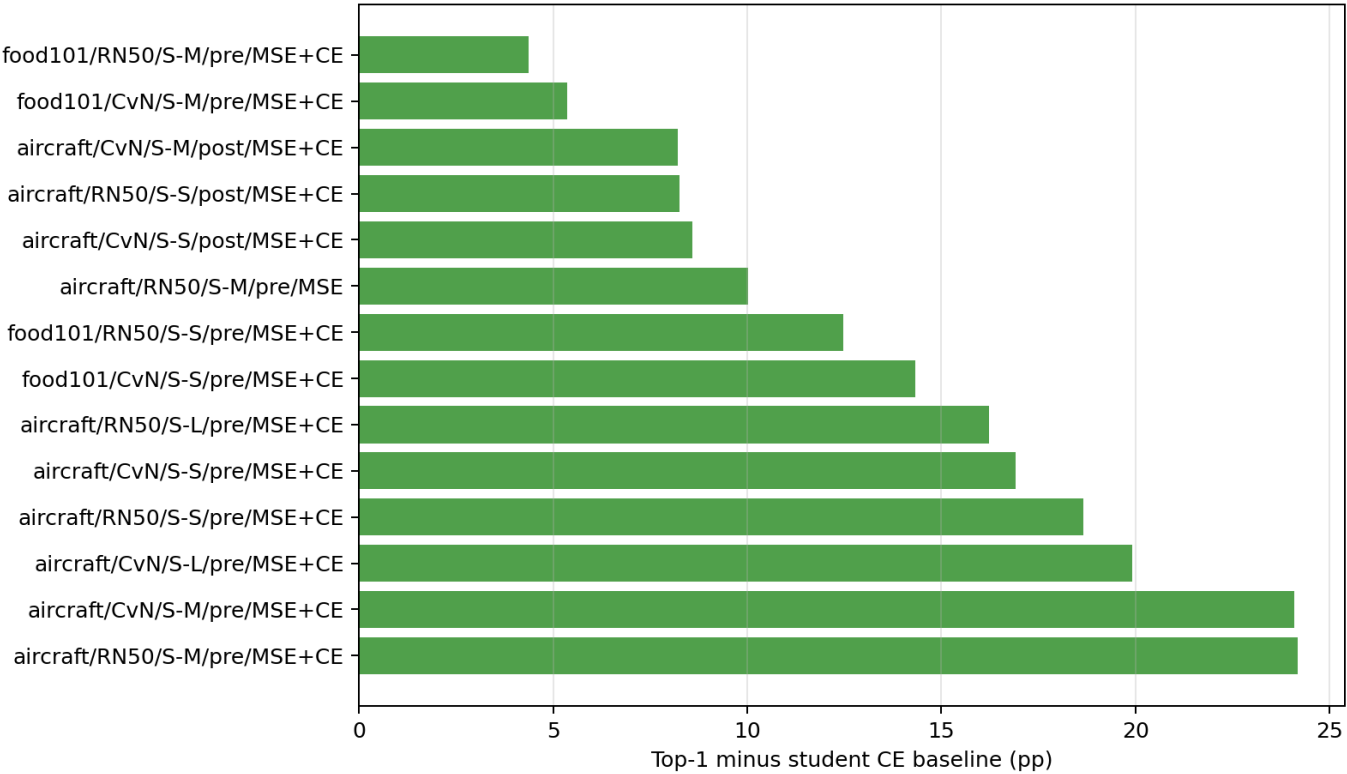
Global student architecture overview using normalized deltas:

student	top1_minus_baseline	top1_minus_teacher	gflops_saved_vs_teacher_pct	params_saved_vs_teacher_pct
student_s	3.775	-41.452	75.810	93.426
student_m	-1.072	-32.163	79.283	91.865
student_l	-9.439	-25.186	76.208	86.442

Global loss overview using normalized deltas:

loss_name	top1_minus_baseline	top1_minus_teacher	gflops_saved_vs_teacher_pct	params_saved_vs_teacher_pct
mse_ce	1.188	-29.500	77.101	90.578
mse	-5.678	-36.366	77.101	90.578

Top distilled students: improvement over matching CE baseline



Global overview: Accuracy (Top-1) vs Compute

